

How the system works

Bravura Electrical Department app

A full guide: what it does, who does what, how work flows, and how it runs, is kept safe and is released.

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1. What the system is

The Electrical Department app is the one place where the electrical team runs its work at every Bravura site (Harare, Kamativi, Manhizi and Selous). It replaces paper job cards, tool issue forms, store cards, WhatsApp groups and spreadsheets with one app that works on Android phones and in any web browser.

What it covers

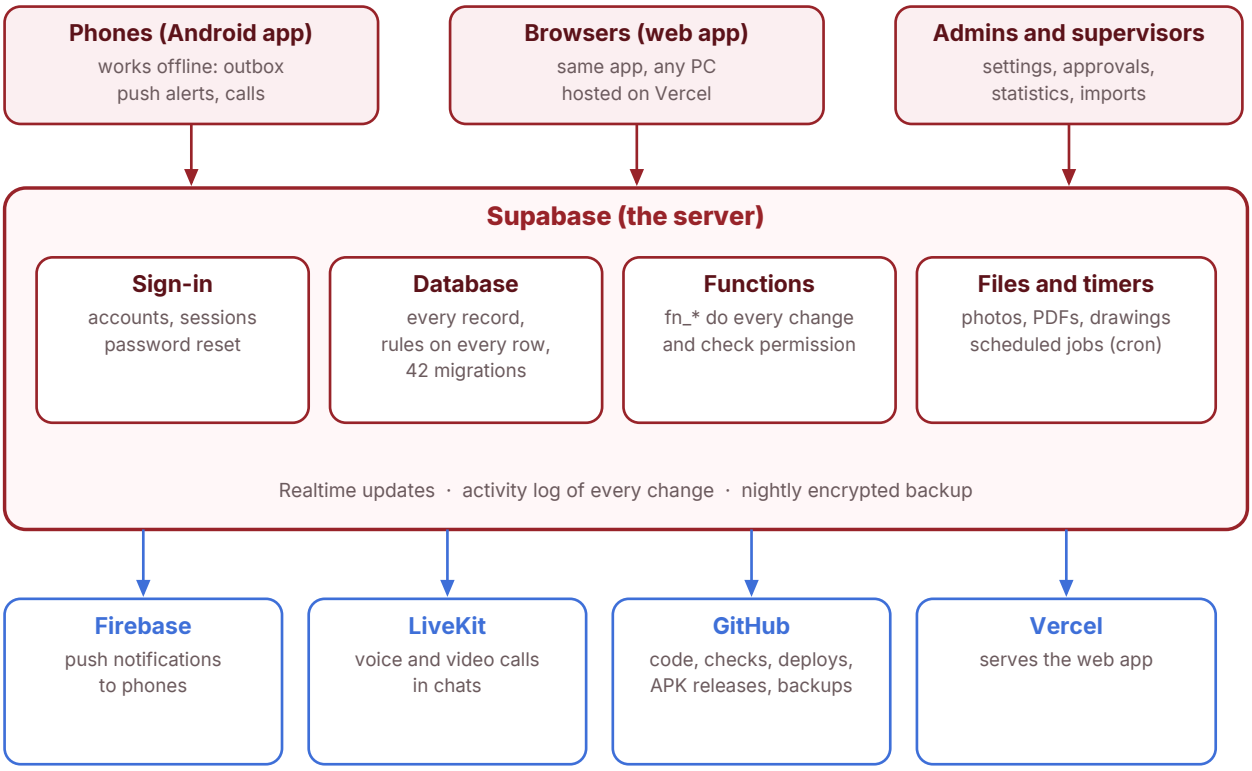
Area	What people do there
Job cards	Report faults, register, assess, assign, do and verify the work, with time, safety checks and photos; requests from other departments.
Tools and equipment	The tools register, signed issue forms, calibration certificates, repairs, who has what; the asset register and planned maintenance.
Inventory and materials	Stock on hand, receiving, issuing, counts; material requests, purchases and project bills of quantities (BOQ).
Projects and planning	Projects with progress, the calendar of everything due, the weekly roster with standby.
Safety and records	SOPs and drawings with read-and-sign, daily checklists (rounds), daily reports with supervisor approval.
Communication	Team and direct chats, voice and video calls, meetings with minutes, notifications and personal reminders.
Management	Statistics for jobs, tools and inventory; users, roles and permissions; activity log; error reports; backups.

Principles it is built on

- **One team, every site.** Everyone can see every site, and everyone's role applies at all sites. The app opens on your home site; the site chip at the top switches, and All my sites lists every site together.
- **Works without signal.** Most work done on a phone is kept on the phone and sent when there is signal again. Nothing waiting is lost when the app closes.
- **Every record has a code** (J-, T-, S-, I-, P-, MR-, PO-, R-, M-, TI-, EL-JC-, EL-SR-) that is never reused, so it can be quoted in chat, on paper and in reports.
- **Nothing is deleted.** Records are archived, and every change is written to the activity log with who and when.
- **The server checks everything.** The app hides buttons people cannot use, but the database itself refuses any change a person is not allowed to make.

2. How it is built

There is one app, written in Flutter, built twice: as an Android app (APK) and as a web app. Both talk to the same server, Supabase, which holds the data, the files and the rules.



Part	Role	Notes
Flutter app	Screens on phones and browsers	Code in app/. Keeps a local database (drift) for offline use and an outbox of changes waiting to send.
Supabase database	All records and the rules	PostgreSQL. 42 numbered migrations in supabase/migrations; every table has row-level security; changes go through fn_* functions.
Supabase Auth	Sign-in	Email and password; admins create accounts and reset passwords.
Supabase Storage	Files	Photos, documents, certificates, signatures, chat files, in private buckets, opened with links that expire.
Scheduled jobs	Timers on the server	Escalations every 10 minutes, reminders every minute, tool and checklist reminders, daily report build, standby reminders.
Edge functions	Server code	Push sending, call tokens (LiveKit), user administration.
Firebase	Push notifications	Alerts reach phones even when the app is closed.
LiveKit Cloud	Calls	Voice and video calls inside any chat.
Vercel	Web hosting	electrical-department-lake.vercel.app, deployed on every change to main.
GitHub	Code and automation	Checks every change, deploys the database and the web app, builds the APK, runs the nightly backup and browser tests.

3. Sites, people, roles and permissions

Sites and areas

Each site (Harare HRE, Kamativi KAM, Manhizi MAN, Selous SEL) has areas (for example "Plant – MCC room"). Records belong to a site, and record codes count up per site, so Harare and Kamativi can each have an S-0001. Admins add sites and areas in Settings → Admin → Sites and areas.

One team

- Every person's roles apply at **every site**. A Harare technician can work a Kamativi job card, do its checklists and sign its SOPs. Alerts come from every site, and everyone is in every site's group chat.
- An admin can still limit a role to one site in Settings → Admin → People, if ever needed.
- Viewing permissions (job cards, tools, inventory, documents, equipment, reports, statistics) count at every site anyway, for anyone who holds them at one site.

Roles

A person has one or more roles. A role is a set of permissions. The table shows what each role may do by default; admins change it in Settings → Admin → Roles and permissions.

Role	What it is for	Main permissions
Technician	Electricians doing the work	See and register job cards, work on job cards assigned to them, do checklists, request materials, see tools, inventory and documents.
Supervisor	Running the team	Everything on job cards (assess, assign, review, cancel, reopen, escalate), approve material requests and daily reports, manage documents, checklists, projects, the roster and meetings, statistics.
Storekeeper	The store and tools	Issue and receive tools, receive, issue, adjust and count inventory, purchases.
Department Manager	Oversight	See job cards, reopen and escalate, reports, statistics, the activity log.
Requester	Other departments	Send work requests to the electrical department and follow them.
Admin	Running the app	Every permission: people, roles, sites, settings, error reports.

> The app shows only the buttons a person may use, but the real check is on the server: every change goes through a database function that checks the permission at that site. A modified app cannot get round it.

4. Finding your way in the app

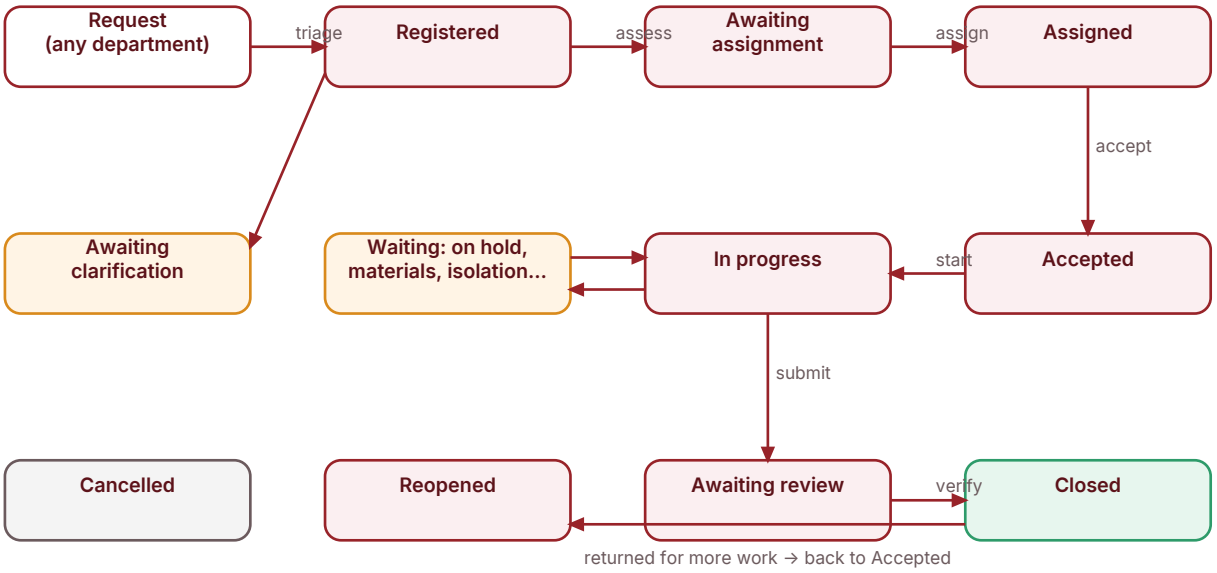
Place	What is there
Home	Greeting, Needs you (critical, overdue, waiting for you), the day's numbers for the site, shortcuts.
Jobs	The job card list with status chips, search and filters; My jobs; statistics.
Chat	Group and direct chats, calls, meetings.
Search	The search button on every tab finds job cards, equipment, tools, documents, inventory, projects, requests and purchases by code or name.
Menu	Every other screen: Requests, Checklists, Daily report, Equipment & tools, Inventory, Materials, Projects, Calendar, Roster, Documents, Reminders, Notifications, statistics.
+ New	The quick-create sheet: report a fault (job card) or request work, my jobs, today's report update, a checklist, request materials, issue tools, a reminder.
Site chip	At the top: the site you are working in; tap to switch.
Settings	Profile, appearance (dark liquid glass by default, light available), notifications, Waiting to send, reload, and for admins the Admin section.

On a phone the bottom bar scrolls sideways to every screen; on a computer the same screens are in the menu on the left. Pushed screens have a back arrow at the top left.

5. Job cards: from fault to closure

Job cards are the department's one job system. Each has a code like EL-JC-2026-0042, a priority, a deadline, a team and a full history.

The stages



Reopened goes back to Assigned or Accepted. Any open stage can be cancelled with a reason.
Time spent waiting (amber) does not count against the deadline.

Stage	What happens	Who
Request	Another department sends a request (EL-SR-…) with photos. The electrical team triages it: registers it as a job card, returns it for more detail, or rejects it.	Requester; triage by technicians and supervisors
Registered	A job card is opened directly (+ New → Report a fault), from a request, from a failed checklist item, from planned maintenance or as a follow-up.	Anyone with register permission
Assess	The supervisor checks the job is valid, sets type, severity, safety risk and priority, or asks the reporter for clarification.	Supervisor
Assign	A responsible person and a team are set; they are notified.	Supervisor
Accept and start	The technician accepts, does the safety check (when required), and starts the work timer.	The assigned technician
Work	Work-log entries (diagnosis, actions, readings), photos and files, tools and materials used, comments with @mentions. Work can be paused for a reason: on hold, waiting for materials, isolation, a specialist or production clearance.	The team

Stage	What happens	Who
Submit	The technician submits the work for review with a summary.	The responsible person
Verify	The reviewer approves (closes it), approves with a follow-up job card, returns it for more work, or asks a question.	Supervisor (not the same person, unless self-approval is allowed)
Closed	Closure freezes the metrics: response time, working time, waiting time, met or missed deadline. A closed job card can be reopened with a reason.	—

Priorities and deadlines

Priority	Meaning	Respond within	Finish within
P1 Emergency	Immediate safety hazard, critical supply failure	15 minutes	4 hours
P2 High	Significant failure, production affected	1 hour	24 hours
P3 Medium	Non-critical fault, limited impact	4 hours	3 days
P4 Low	Minor repairs, inspections, improvements	1 day	14 days

Admins can change these targets in Settings → Admin → Job card settings. Time spent in a waiting state moves the deadline, so a job is not counted late while it waits for materials.

Escalation

Every 10 minutes the server checks open job cards against nine escalation rules (for example: a critical job not yet assigned, a job close to or past its deadline, a job on hold too long) and notifies the right people once per rule and deadline. Supervisors and managers can also escalate a job card by hand.

Linked to everything else

- **Tools** on a job card: add the tools the job needs, see who has each, issue them on a form.
- **Documents**: attach the SOP or drawing for the job.
- **Materials**: request materials for the job; stock issued to it is listed.
- **Projects**: a job card can belong to a project, whose progress counts it.
- **Chat**: type # to share a job card in a chat as a card that opens it.

6. Tools and equipment

The tools register

Every tool has a T- code, kind, serial number, condition (good, needs repair, damaged, lost) and, for meters and testers, a calibration due date. The register shows what is in the store, out, overdue back and due for calibration.

Issuing tools on a form

- The store issues one or more tools to a person (an app user or anyone, for example a contractor) on a **tool issue form** (TI-), with company, team lead, place, return date, the condition of each tool, and **three signatures**: issuer, recipient and team lead.
- Taking tools back records the condition of each and three signatures again; part of a form can come back at a time. The form prints as a PDF laid out like the company's paper form.
- Tools already out, damaged, lost or with expired calibration cannot be issued.

Repairs, calibration, who has what

- A tool returned **damaged or needing repair** opens a repair job card automatically; when that job card is closed the tool is back to good and the store is told.
- **Calibration certificates** (PDF or photo) are uploaded with their dates; the newest sets the due date. Reminders go out 7 days before.
- **Who has what** lists everyone holding tools now, with their company and what is overdue.
- **Tool stats**: losses, late returns, damage with its repair job card, calibration, and a summary per person, for 7 days to a year, one site or all.
- Daily at 07:15 Harare the server reminds holders of overdue tools and the store of calibration due.

Equipment and planned maintenance

The asset register lists plant and equipment with tag, category, location, ratings, criticality and status. **Planned maintenance** plans (daily, weekly, monthly) raise preventive job cards automatically on their dates.

7. Inventory

Inventory (formerly the store) holds electrical stock per site. Each item has an I- code, unit, category, bin and an alert below level. Stock on hand only changes through movements, so it always equals the sum of its history and never goes below zero.

Movement	Who	What it does
Receive	Store	Stock in (a delivery), with a supplier or delivery note.
Issue	Store	Stock out to a job card, a project or a person; refused if not enough.
Take back	Store	Unused stock returned from a job card or a person.
Adjust	Store	A correction with a required reason (damaged, found, written off).
Count	Store	Count the shelves by bin, offline too; a count in progress is kept if you leave. Only items that differ are corrected, as of when they were counted (later issues stay). Past counts lists every count and its corrections.

- When an item falls below its level, the store and supervisors are alerted.
- Stock on approved material requests not yet issued is **reserved**; each item shows how much is reserved and how much is free.
- The list is saved on the phone; without signal it shows the saved stock with the time it was saved. Movements made offline are sent later.
- **Import from Excel**: load items with opening stock from a spreadsheet.
- **Inventory stats**: what was used most, stock per project, job cards using the most, corrections, what is low.

8. Materials, purchases and bills of quantities

Material requests (MR-)

- Anyone in the team requests materials for a job card, a project or the site: items from inventory, or things not stocked, typed in.
- A supervisor approves, or rejects with a reason. The requester is told either way.
- The store **issues** what is in inventory (to the request's job card and project) and **orders** what is missing.

Purchases (PO-)

- A purchase has a supplier, the company PO number, expected date, lines and prices. It can be filled in from a material request in one tap.
- Status: to order → ordered → delivered (or cancelled). **Receive delivery** puts items into inventory; something not stocked counts as handed over on its request. Whoever asked is told it has arrived.
- Prices are seen only by the store and supervisors.

Bill of quantities (BOQ)

A project's BOQ is imported from Excel (or pasted). Each line shows how much was issued, is requested, is on order and is in inventory, and what is still short. **Request the shortfall** turns everything missing into one material request.

9. Projects, calendar and roster

Projects (P-)

A project groups job cards, materials and a BOQ, with a target date, a leader and a team. Its progress ring shows job cards closed out of all (not cancelled). Team members are notified when added.

Calendar

A month view with a coloured dot for each kind of thing due: job cards by their deadline, meetings, tools due back, calibration, planned maintenance, project deadlines, your own shifts and standby, and your reminders. Tap a day for its list; tap an entry to open it.

Roster and standby

- Supervisors set each person's day for the week: day, night, off or leave, plus standby (on call).
- Publishing the week tells everyone on it. People on standby get a reminder at 16:00 the day before.
- Everyone sees My shifts at the top of the roster.

10. Documents, checklists and daily reports

Documents (S-)

- SOPs, drawings, manuals and forms per site, in folders, as a grid or list, with filters (to sign, recent, added by me, type) and coloured file-type badges.
- Each document keeps every version. A document marked everyone must read and sign asks people to sign the current version; a new version asks again. Supervisors see who has signed.
- Upload several files at once (or drop them onto the page on a PC); drag a document onto a folder to file it; share in a chat; copy a 7-day link for someone without the app; move, edit, archive.

Checklists (daily rounds)

- Supervisors build templates per area: tick items, readings with limits, text, required photos; daily, weekly or monthly.
- Technicians do the round on the phone, even offline. A failed tick or a reading out of limits **raises a job card**.
- At 15:00 the assignee is reminded of rounds due today; at 19:00 supervisors hear what was missed.

Daily reports (R-)

- Each site has a daily report built automatically (19:10 every night): job cards opened and closed, checklists done and missed, tools out. Everyone adds their own update with photos.
- A supervisor approves the report with a comment; it exports as a PDF.
- The **reports archive** lists every report and searches through what people wrote.

11. Chat, calls and meetings

- **Chats**: every site has a group, the department has one, and anyone can start a direct chat. Messages, photos, voice notes, files, replies, reactions, @mentions, pins; documents and records (type #) shared as cards that open them. Messages sent offline go when there is signal.
- **Calls**: voice or video in any chat, with screen sharing and speaker on phones; incoming calls ring the phone.
- **Meetings (M-)**: scheduled from a group chat with an agenda; everyone in the chat is invited and reminded 15 minutes before, and joins through the chat's call. Minutes hold a summary, decisions and actions with owners and dates. Automatic transcription waits for its service key.

12. Notifications and reminders

- Notifications appear in the app (the bell) and as push alerts on phones and browsers that allow them. Each opens the record it is about.
- Examples: a job card assigned to you, a mention, a job going overdue or escalated, something to approve, low stock, a meeting starting, a roster published, a delivery arrived, tools overdue.
- **Personal reminders**: "Remind me" on a job card or tool, or from + New: in 1 hour, later today, tomorrow 08:00 or any time. They work offline and arrive as a notification.

13. Statistics

Screen	Shows
Job statistics	Trends, job cards by priority, type, area, equipment and technician, overdue ageing, response and closure times.
Tool stats	Losses, late returns, damage and repairs, calibration, per person and company.
Inventory stats	Most used items, stock per project, job cards using the most stock, corrections, low stock, open requests and purchases, project status.
Equipment	Each asset's faults in 30 and 90 days, time out of service, and a Repeat faults sign; Job statistics lists repeat faults and downtime.
Home	The day's open, critical, overdue and closed-today counts, and what needs you.

Each works for one site or All my sites, over a chosen period.

14. Working without signal

Mine sites often have weak or no signal, so the app is built to keep working.

What works offline

- Viewing what was last loaded: job cards, tools, equipment, people, inventory, checklists.
- Settings → Waiting to send shows when something last reached the server and how long the oldest unsent change has waited.
- Doing work: job card notes, work-log entries and photos, checklists, daily report updates, tool issue forms and returns, inventory movements and counts, material requests, projects, rosters, document

uploads, chat messages, reminders.

How changes reach the server

- Each change goes into the phone's **outbox** with its own id, files first, then changes in the order they were made. The same change sent twice is saved once.
- No signal is never a failure: the app tries again after 30 seconds, then waits longer each time (up to 10 minutes), and goes straight away when the app comes back to the foreground or someone taps Send now.
- Only the server refusing a change marks it not saved, for example a note on a job card that someone else cancelled meanwhile. It then shows in the banner and in Settings → Waiting to send with **Try again** and **Discard**.
- A change whose file the server refused waits until the file is tried again or discarded, so no record points at a missing photo.
- **Signing out** with work not yet sent asks first: wait, see what is waiting, or sign out and delete it. Nothing unsent is deleted silently.
- Up to 300 MB of files can wait on a phone. Status changes and work timers need signal, so the server's times stay true.

15. Administration

Screen (Settings → Admin)	What admins do
People	Create accounts, activate or deactivate, roles (at all sites or one), reset passwords.
Roles and permissions	Switch what each role may do. The Admin role always keeps user management.
Sites and areas	Add and rename sites and their areas.
Job card settings	Priorities and targets, numbering, categories, departments, escalation rules and contacts, safety checklist items and when the safety check is required.
Activity log	Every change to every record: who, what, when.
App errors	Errors that happened on people's phones and browsers, grouped, with how often, for how many people, the app version and the screen; mark fixed.

Imports from Excel (inventory items, tools, BOQs) accept .xlsx or CSV files whose first row names the columns; the app works out which column is which.

16. Keeping the data safe

- **Access**: private by default. Every table has row-level security; every change goes through a function that checks the person's permission at that site. Files open only through links that expire.
- **History**: nothing is deleted; records are archived and every change is in the activity log.
- **Backups**: every night at 00:20 Harare the live database is copied, restored into a test database and compared table by table, row counts and a checksum of every value (so the backup is known to work), encrypted with a passphrase and kept 90 days. Photos and documents are copied, encrypted, to storage of your choice once set up (docs/backup-files.md).
- **Permissions audit**: nobody signed out can touch any table, nothing can be written past the app's checks, and the last admin cannot be removed; tests keep it so.
- **Secrets** (keys, passwords) live only in GitHub and Supabase settings, never in the code.
- **Privacy**: the privacy policy is published at /privacy.html on the web app.

17. How changes are made and released

Step	What happens
1. A change	Code is changed in the repository (GitHub). Database changes are a new numbered migration with a test; old migrations are never edited.
2. Checks	GitHub checks every push: code format, analysis, app tests (82), and every database test against a throwaway copy of the database (688 checks).
3. Deploy	On main: the migrations are applied to the live database, and the web app is built and published on Vercel within minutes.
4. Android	Every push builds an APK. A release (run by hand with a version like v0.1.3) builds the signed APK and the Play Store bundle, attaches them to a GitHub release and sends them to testers through Firebase App Distribution.
5. Nightly	Browser tests run the main flows every night; the backup runs at 00:20.

Phones get new features only with a new APK release; the web app updates on every deploy.

18. Record codes

Code	Record	Code	Record
EL-JC-YYYY-NNNN	Job card	EL-SR-YYYY-NNNN	Request from another department
T-	Tool	TI-	Tool issue form
I-	Inventory item	MR-	Material request
PO-	Purchase	P-	Project
S-	Document (SOP, drawing…)	R-	Daily report
M-	Meeting	J-	Old jobs board job (read only)

19. Still waiting on people

Everything planned has been built. These steps need a person with a phone, an account or a secret:

Item	Why it matters
Release to phones	Phones are on 27 October's version and miss three weeks of work, including the offline and sign-out fixes.
BACKUP_PASSPHRASE secret in GitHub	Without it the nightly backup stops rather than keep an unencrypted copy.
Firebase App Distribution role	So testers receive new APKs automatically.
Transcription key (Groq)	For automatic meeting transcription.
Phone testing	Run docs/offline-test-plan.md on an Android phone.
Play Store account (optional)	A one-time 25 US dollars; the bundle, policy and listing are ready.

More detail for developers is in API.md (every table, function and rule), CHANGELOG.md (what changed when) and docs/ in the repository.